



Exploring Perceived Interactions between EQ-5D-5L and Bolt-ons Using Composite Time-Tradeoff Valuations: A Qualitative Study

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Objectives. This study qualitatively explored how bolt-ons affect the perception of EQ-5D-5L core dimensions in a valuation context. **Methods.** Sixty Indonesian adults (aged 20–67 y, 50% female) each valued 10 health states using composite time tradeoff (cTTO). States were presented in either forward (EQ-5D-5L, then with 1 bolt-on, then 2) or backward (reversed sequence) order. Participants were assigned to 1 of 3 bolt-on dyads: vision and tiredness, cognition and social relationships, or skin irritation and self-confidence. In semistructured qualitative interviews, respondents described how adding or removing bolt-ons changed the perceived importance of the 5 core dimensions. We classified these changes as related either to measurement or to valuation, with the latter further categorized as a relative or absolute shift in importance. **Results.** Cognition and vision generated the most shifts in perceived importance in the forward and backward groups, respectively. Regardless of ordering group, most shifts occurred between the EQ-5D-5L alone and the version with 1 bolt-on (first presented in the dyad), with significantly fewer shifts observed between the 1-bolt-on and 2-bolt-on states. In the forward group, most shifts were classified as measurement (56%) or relative preference (29%), while the reverse was true in the backward group: relative preference (53%) or measurement (31%). Absolute preference was least common across both groups. **Conclusions.** This is the first study to explore how individuals reason when valuing EQ-5D-5L + bolt-on health states. Our findings suggest that interactions between dimensions are complex and may be influenced by presentation order. Further qualitative research should directly investigate absolute preferential reasoning.

Highlights

- No studies have qualitatively explored how individuals value EQ-5D health states with bolt-ons. Understanding how bolt-ons influence reasoning and interact with core dimensions is crucial for informing valuation methods and modeling strategies.
- Our findings show that bolt-ons can alter how participants perceive the importance of EQ-5D-5L dimensions, although changes are mostly not preference driven. Participants often rely on accessible reasoning, such as conceptual associations between dimensions. Effects vary by the bolt-on used and presentation order.
- Interactions between bolt-ons and core dimensions complicate efforts to develop robust valuation approaches. Future qualitative studies should aim to capture preference-based reasoning, while quantitative work is needed to disentangle preferential from nonpreferential effects.

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The EQ-5D is the most widely used standardized generic instrument for measuring health-related quality of life (HRQoL) and generating utility values for economic evaluation.¹ Its concise design, which includes 5 core dimensions (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression), support universal applicability and facilitate comparison across populations and studies. However, this brevity may limit its ability to fully capture aspects relevant to specific populations or conditions. “Bolt-ons” are additional dimensions that can be attached to the standard EQ-5D to account for aspects not covered by the 5 core dimensions.² Bolt-ons have been shown to enhance the measurement properties of the EQ-5D, including reducing ceiling effects and improving content validity, discriminatory power, and responsiveness.^{3–12}

Adding bolt-ons to the EQ-5D can substantially influence health utility values.¹³ Previous studies have examined the effects of various bolt-ons (e.g., energy/tiredness, cognition) on health preferences,^{14–17} and experimental value sets have been developed for several, including vision, psoriasis, and breathing problems.^{13,18,19} Various valuation methods have been applied in this context, including variants of time tradeoff (TTO), discrete choice experiments, person tradeoff, and analytical hierarchy process.^{20–24}

More recently, alternative approaches such as the Kaizen task and Online elicitation of Personal Utility Functions have also been tested.^{25,26} While no consensus exists on a standardized valuation protocol for bolt-ons, one promising approach is adapting national value sets for bolt-on integration, such as through the “scaling factor model.”

The scaling factor model allows the coefficients of an existing value set to adjust proportionally (typically expected to shrink) when a bolt-on is added, while preserving the accuracy of health preferences in the utility values.^{27,28} This approach assumes that the relative importance of core dimensions and their levels, in terms of disutilities, remains constant between core and bolt-on value sets. A central motivation behind this approach is that introducing a bolt-on alters how individuals interpret the core EQ-5D dimensions, which subsequently affects disutilities. By making the bolt-on explicit, the model assumes individuals no longer infer its content from the core dimensions, potentially diminishing the importance weight assigned to them. However, it remains unclear whether people organically make such inferences when imagining EQ-5D health states, that is, whether they mentally include additional HRQoL areas such as memory, sleep problems, or blindness. For instance, some may assume that a health state such as “21111” (slight problems with mobility and no problems in self-care, usual activities, pain/discomfort, and anxiety/depression) implies normal vision or that “55555” (extreme problems in all core dimensions) includes extreme sleep problems.

Although the quantitative literature on bolt-on valuation is expanding—introducing new elicitation methods and modeling strategies—there is yet to be qualitative insight into how individuals cognitively process EQ-5D + bolt-on health states. Therefore, this study aims to explore how individuals value EQ-5D-5L health states with and without bolt-ons and to understand the potential interactions or perceived connections between bolt-ons and the 5 core dimensions.

Methods

Study Design

This qualitative study comprised 3 components: composite TTO (cTTO) valuation tasks, a semistructured

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qualitative interview, and a sociodemographic questionnaire. The cTTO tasks served to prompt participant reflection during the interview, while the questionnaire collected background characteristics for contextual interpretation.

To examine whether bolt-ons influenced how individuals interpret core EQ-5D-5L dimensions (potentially affecting valuation outcomes such as years traded in cTTO), we applied 2 experimental presentation orders. In the forward stepwise order, participants began with EQ-5D-5L and sequentially added bolt-ons. In the backward stepwise order, they started with EQ-5D-5L + bolt-ons and sequentially removed the bolt-ons. This design enabled exploration of how presentation direction may shape the interpretation of core dimensions. As interpretation may differ by bolt-on, both for the forward and backward order, 3 study arms were constructed using distinct bolt-on dyads. The first arm (A1) included vision and tiredness, the second (A2) included cognition and social relationships, and the third (A3) featured skin irritation and self-confidence. We hypothesized that the conceptually related bolt-ons in A2 would produce greater interpretive interactions than the less related bolt-ons in A1. Meanwhile, A3 included a dyad that had been previously developed.¹⁹ Details of the health descriptors and bolt-ons are provided in Supplementary Material 1.

The cTTO tasks were conducted using a modified EuroQol Portable Valuation Technology (EQ-PVT), a macro-enabled Microsoft PowerPoint program that serves as a compact version of the EQ-VT software.²⁹ The qualitative interview was the core component of the study, designed to explore participants' thought processes during cTTO valuation. As a concluding step, a sociodemographic questionnaire was administered, covering education, income, residential setting, and employment, marital, and parental status.

Six pilot interviews were conducted prior to data collection to test the feasibility of the cTTO tasks in EQ-PVT, the interview guide, and the questionnaire. Based on feedback, minor wording adjustments were made to the guide to reflect commonly used colloquial terms. All tasks and interviews were conducted online via Zoom in the Indonesian language.

Participants

Between November and December 2024, Indonesian adults from the general population were purposively recruited through social media (WhatsApp and Facebook), which was extended through snowball sampling. Sampling aimed to reflect the age and sex distribution of the adult population in Indonesia. Participants were

evenly randomized to either the forward or backward ordering group and subsequently to 1 of the 3 study arms, resulting in 6 subarms. A target of 60 interviews (10 per subarm) was set to reach thematic saturation, based on similar studies.^{30–32} Inclusion criteria included age ≥ 17 y, fluency in Indonesian, residence in urban or rural Java, no cognitive impairment, and access to a personal computer with adequate internet connection in a private space suitable for interviews. Given the complexity and reflective nature of the interviews, we intentionally oversampled individuals with higher education. Participation required informed consent, which covered video recording and the use of pseudonymized data for research purposes. Participant identifiers indicated the study arm (A1, A2, or A3), ordering group (F = forward, B = backward), and number within that subarm (e.g., A2_F_04: participant 04 from Arm 2, forward order). All participants received IDR 200,000 compensation ($\approx$ \$12.60).

cTTO Valuation Tasks Using EQ-PVT

To reduce participant burden and avoid unrealistic combinations of dimension levels, we included only level 2 ("slight") and level 4 ("severe") problems in the EQ-5D-5L (+ bolt-on) health states. In the forward group, participants began with EQ-5D-5L, followed by EQ-5D-5L + first bolt-on (A1: vision; A2: cognition; A3: skin irritation) and, finally, EQ-5D-5L + bolt-on dyad. In the backward group, the sequence was reversed, starting with the EQ-5D-5L + bolt-on dyad, then EQ-5D-5L + first bolt-on (after removing the second: A1: tiredness; A2: social relationships; A3: self-confidence), and finally EQ-5D-5L.

All participants began with a set of warm-up questions (age, sex, and experience with serious illness), followed by the EQ-5D-5L descriptive system and EQ VAS. In the forward group, participants completed 2 example health states ("being in a wheelchair" and "worse than being in a wheelchair") and 1 practice EQ-5D-5L health state (31232). They then 1) valued 2 EQ-5D-5L states: 22222 and 44444; 2) completed the first bolt-on item to self-report their own health; 3) valued 4 EQ-5D-5L + first bolt-on states: 22222 + 2, 22222 + 4, 44444 + 2, and 44444 + 4; 4) completed the second bolt-on item self-report; and 5) valued 4 EQ-5D-5L + bolt-on dyad states: 22222 + 2 + 2, 22222 + 2 + 4, 44444 + 4 + 2, and 44444 + 4 + 4. The order of health states was randomized within 1), 3), and 5), with debriefing questions about participants' decision making asked after each step.

In the backward order group, participants first completed the bolt-on dyad to self-report their own health, followed by 2 example health states and 1 practice EQ-5D-5L + bolt-on dyad health state (31232 + 3 + 1). They then valued 1) 4 EQ-5D-5L + bolt-on dyad states: 22222 + 2 + 2, 22222 + 2 + 4, 44444 + 4 + 2, and 44444 + 4 + 4; 2) 4 EQ-5D-5L + first bolt-on states: 22222 + 2, 22222 + 4, 44444 + 2, and 44444 + 4; and 3) 2 EQ-5D-5L states: 22222 and 44444. Health states were randomized within each step. Debriefing questions, identical to those used in the forward group, were asked after each step.

All interviews were conducted by S.P., who had prior training in the EQ-VT protocol and experience in online qualitative interviewing.³³ For familiarization with the EQ-PVT tasks, 5 practice interviews were conducted with a convenience sample prior to the study's pilot phase. Two other researchers, both with extensive experience in EQ-5D valuation studies, reviewed the interviews after every 10 sessions, following the EuroQol standardized quality control process.³⁴ This included evaluating whether sufficient time was spent on the cTTO tasks, thoroughly explaining the 2 wheelchair examples, and using the lead-time TTO.

Qualitative Interview

Following the EQ-PVT valuation tasks, semistructured interviews were conducted using a topic guide developed by the research team. In the forward group, questions explored whether adding bolt-ons influenced how participants considered the core dimensions. In the backward group, questions focused on whether removing bolt-ons led participants to infer the level of problems in the removed dimension based on the remaining ones. In both groups, participants were asked whether the addition or removal of bolt-ons changed the perceived importance of the core dimensions. The full topic guide, including additional questions and probes, is provided in Appendix Material 2.

Qualitative Data Analysis

All recorded interviews were manually transcribed in Indonesian by 2 research assistants and verified by the 2 Indonesian members of the research team. These researchers reviewed transcripts multiple times to extract responses related to changes in the perceived importance of EQ-5D-5L core dimensions following bolt-on addition or removal. One researcher translated the excerpts into English and aggregated the reported changes across the 5 core dimensions to capture overall directional shifts (increase [↑] or decrease [↓]). For example, removing self-

confidence may have led participants to place greater importance on anxiety/depression (backward order, $k = 1$ shift), while adding social relationships may have caused mobility to be considered less important (forward order, $k = 1$). Each participant could contribute multiple shifts. To interpret these shifts, we used the Framework method.³⁵ Excerpts were coded to develop an initial analytical framework, refined through team consensus and informed by prior experience in health valuation studies. Thematic saturation was achieved through iterative review of emerging codes across interviews within and across study arms and ordering groups, with no substantively new themes identified after the sixth to seventh interview within subgroups.

The final framework captured 3 types of participant reasoning: 1) "measurement," perceived conceptual links or overlaps between dimensions; 2) "relative preference," comparisons between dimensions; and 3) "absolute preference," reassessment of a dimension's standalone importance due to a bolt-on, implying a potential influence on valuation decisions in cTTO tasks (i.e., willingness to trade fewer or more years). Table 1 presents a conceptual overview of the 3 reasoning types, outlining their underlying cognitive mechanisms, expected impact on cTTO values, and implications for valuation modeling assumptions. The final thematic framework was applied independently by S.P. and M.J. to classify the excerpts, achieving 87.7% agreement, with discrepancies resolved through team consensus. Results were supported by frequency summaries and illustrative quotes, presented separately for the forward and backward groups, and by study arm.

Results

Of 82 individuals reached through the recruitment process, 60 were ultimately included in the study (Figure 1). There were 30 participants in each of the forward and backward groups, with each group consisting of 3 subarms of 10 participants. The median age was 39.5 y (range = 20–67 y), with equal sex distribution (50% female; Table 2). Most participants had tertiary education (81.7%) and were urban based (88.3%). The median interview duration was 53 min.

Forward Group

Perceived impact of bolt-on addition. All forward group participants ($n = 30$) expressed that the addition of bolt-ons made the health states seem more severe. Most reported that the bolt-ons prompted them to reconsider how they interpreted the core dimensions; most

Table 1 Conceptual Mapping of Reasoning Types Derived from the Qualitative Analysis

Reasoning Type	Cognitive Mechanism	Expected Impact of cTTO Values	Implications for Valuation Studies
Measurement	Interpreting one dimension as conceptually related to, or overlapping with, another dimension.	No impact on valuation results (in terms of disutilities of individual dimensions). May render some health states implausible (problems in selected dimensions perceived as likely to occur jointly).	Implications for study designs (levels of overlapping dimensions should not differ substantially). Consequently, it may be more difficult to disentangle the effects of these dimensions due to correlation in the design.
Relative preference	Explicit comparison or prioritization of one dimension over another when forming choices.	No impact on valuation results (in terms of disutilities of individual dimensions). Some preference elicitation methods require prioritization of dimensions (e.g., Kaizen Tasks and Online Elicitation of Personal Utility Functions); in such cases, there may be an impact on preference paths or dimension rankings.	Information about the relative importance of bolt-ons versus core dimensions may be needed to design efficient studies when using discrete choice experiment (for relatively unimportant bolt-ons, it may be difficult to estimate disutility weights; for relatively important ones, they may override other dimensions in driving participants' choices).
Absolute preference	Reevaluating the importance of a dimension due to a bolt-on addition or removal.	There will be a change in the disutilities of individual dimensions.	The disutilities of core EQ-5D dimensions need to be reestimated for bolt-on value sets. Impacts across individual core dimensions may differ; therefore, using a single scaling factor may not be justified.

cTTO, composite time tradeoff.

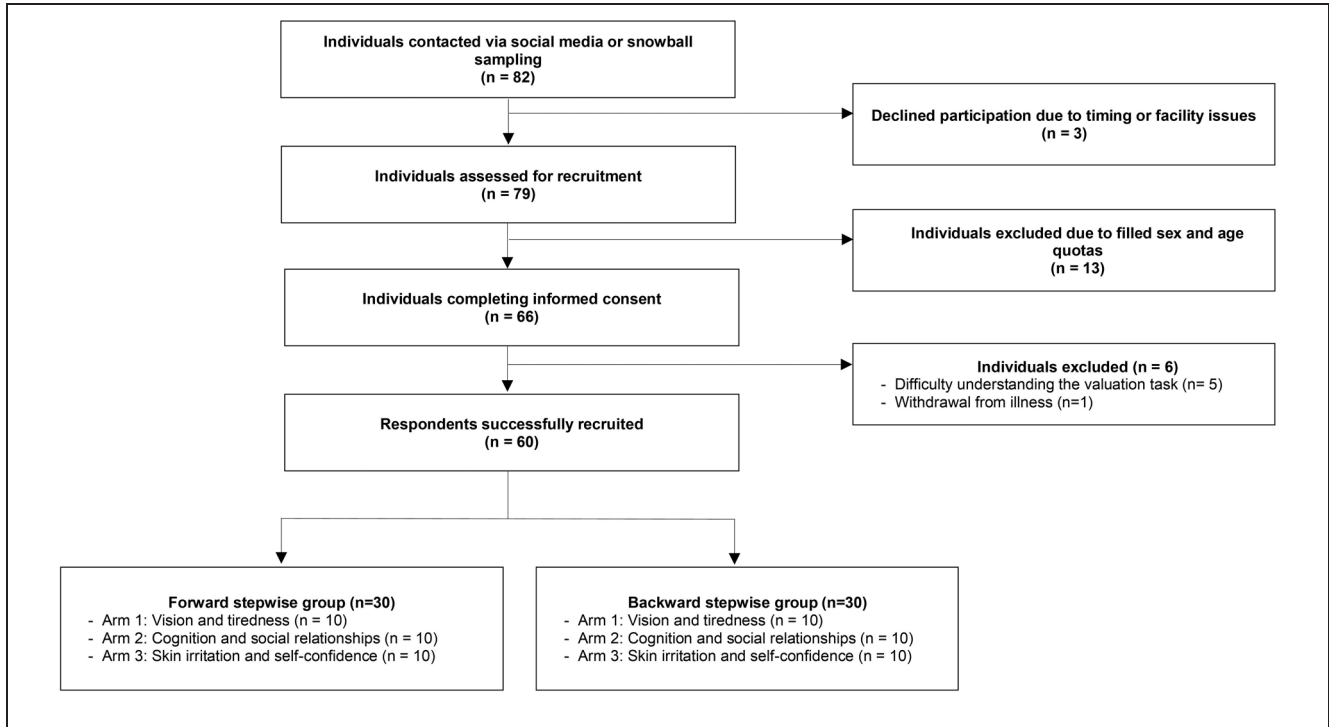
**Figure 1** Participant flow.

Table 2 Characteristics of the Participants

Characteristic	<i>n</i> (%) or Median (Range)
Age (y)	39.5 (20–67)
17–29	16 (26.7%)
30–39	14 (23.3%)
40–49	12 (20.0%)
50–59	9 (15.0%)
60+	9 (15.0%)
Sex	
Male	30 (50.0%)
Female	30 (50.0%)
Highest level of education	
Primary or less	2 (3.3%)
Secondary	9 (15.0%)
Tertiary	49 (81.7%)
Personal net monthly income (IDR) ^a	
Do not want to answer	13 (21.7%)
Up to 5,000,000	7 (11.7%)
5,000,001–10,000,000	5 (8.3%)
10,000,001–15,000,000	7 (11.7%)
15,000,001–20,000,000	6 (10.0%)
>20,000,000	22 (36.7%)
Residential setting	
Urban	53 (88.3%)
Rural	7 (11.7%)
Employment status	
Full-time work	44 (73.3%)
Part-time work	4 (6.7%)
Owns business	6 (10.0%)
Retired	4 (6.7%)
Student	2 (3.3%)
Civil status	
Single	26 (43.3%)
Married	31 (51.7%)
Divorced/widowed	3 (5.0%)
Parent with child(ren) aged <17 y	
Yes	15 (25.0%)
No	45 (75.0%)
EQ-5D-5L profile of “11111” (full health)	25 (41.7%)
EQ VAS ^b	85 (40–100)
≥80	44 (73.3%)
<80	16 (26.7%)
History of physician-diagnosed chronic illness	
Yes	39 (65.0%)
No	21 (35.0%)
Experience with serious illness	
In self	16 (26.7%)
In family	45 (75.0%)
In caring for others	34 (56.7%)
Interview duration ^c (min)	
Forward stepwise (<i>n</i> = 30)	53 (40–69)
Backward stepwise (<i>n</i> = 30)	53 (39–80)

^aIDR 5,000,000 (Indonesian Rupiah) is approximately 309 USD, based on the Bank Indonesia middle exchange rate as of December 31, 2024.

^bHealth group defined as an EQ VAS score of 80 or greater.³⁶

^cExcluding time spent on the final demographics questionnaire.

commonly observed with cognition (*n* = 9; 90%), vision (*n* = 7; 70%), skin irritation, and self-confidence (*n* = 6; 60% each) (Table 3). In addition, cognition was identified as the most important of all dimensions used to define health states, including the core dimensions, by 50% (*n* = 5) of the participants in A2.

Shifts in perceived core dimension importance. As hypothesized, A2 (*k* = 29) elicited more total shifts than A1 (*k* = 24), with A3 (*k* = 27) falling in between (Table 4). The addition of cognition (*k* = 26 shifts) led to the greatest number of decreases in perceived importance across core dimensions, while skin irritation (*k* = 13) and vision (*k* = 12) led to the greatest number of increases. In each arm, the first bolt-on added generated more overall shifts than the second did: vision (*k* = 16) versus tiredness (*k* = 8), cognition (*k* = 27) versus social relationships (*k* = 2), and skin irritation (*k* = 15) versus self-confidence (*k* = 12). Of the overall increases (*k* = 41), most (*k* = 35; 85%) were categorized as occurring in a measurement sense. Meanwhile, among the decreases (*k* = 39), most were categorized as relative preference (*k* = 20; 56%), followed by measurement (*k* = 10; 26%) and absolute preference (*k* = 7; 18%). Notably, 78% of relative preference shifts (*k* = 18) were attributed to cognition, while 58% of absolute preference shifts were attributed to self-confidence (*k* = 7). Across all arms, perceived importance shifted most frequently in the anxiety/depression dimension (*k* = 20), while the remaining core dimensions showed similar frequencies (*k* = 14–16). Below are illustrative excerpts representing each reasoning category (more excerpts in Supplementary Materials 3).

Measurement sense. The addition of skin irritation increased the importance of pain/discomfort:

When something is itchy, it creates a burning sensation which causes both pain and anxiety. (A3_F_07, male, 43)

Interpretation: Itching was interpreted as closely associated with pain/discomfort; considered a specific example of the latter.

Relative preference. The addition of cognition decreased the importance of pain/discomfort:

Before, pain/discomfort was the most important aspect to me, but after the bolt-ons were added, I thought pain would be more manageable than thinking problems. (A2_F_06, male, 63)

Interpretation: The addition of cognition reduced the perceived importance of pain/discomfort for the participant,

Table 3 Perceived Impact of Adding or Removing Bolt-ons

Forward Stepwise Group (<i>n</i> = 30)					Backward Stepwise Group (<i>n</i> = 30)				
Bolt-on Dimension	Bolt-on Changed How Participants Considered the Core Dimensions		Bolt-on Became the Most Important among the 7 Dimensions		Bolt-on Dimension	Still Thought about and Factored in the Bolt-on after Removal		Considered Core Dimensions Differently after Bolt-on Was Removed	
	<i>n</i>	%	<i>n</i>	%		<i>n</i>	%	<i>n</i>	%
Arm 1 (<i>n</i> = 10)					Arm 1 (<i>n</i> = 10)				
Vision ^a	7	70.0%	2	20.0%	Vision ^a	4	40.0%	8	80.0%
Tiredness ^b	3	30.0%	0	0.0%	Tiredness ^b	3	30.0%	3	30.0%
Arm 2 (<i>n</i> = 10)					Arm 2 (<i>n</i> = 10)				
Cognition ^a	9	90.0%	5	50.0%	Cognition ^a	5	50.0%	7	70.0%
Social relationships ^b	3	30.0%	0	0.0%	Social relationships ^b	4	40.0%	5	50.0%
Arm 3 (<i>n</i> = 10)					Arm 3 (<i>n</i> = 10)				
Skin irritation ^a	6	60.0%	0	0.0%	Skin irritation ^a	2	20.0%	4	40.0%
Self-confidence ^b	6	60.0%	1	10.0%	Self-confidence ^b	1	10.0%	4	40.0%

^aFirst bolt-on added in the “forward stepwise” group or second bolt-on removed in the “backward” stepwise group.

^bSecond bolt-on added in the “forward stepwise” group or first bolt-on removed in the “backward stepwise” group.

as thinking problems were viewed as more difficult to manage.

Absolute preference. The addition of skin irritation increasing the importance of self-care:

If I had skin problems, I'd pay more attention to my personal hygiene and also dress to cover up the affected areas. (A3_F_04, male, 23)

Interpretation: Being able to hide, manage, or alleviate the consequences of skin problems through hygiene or dressing appropriately requires no problems with the self-care dimension. Therefore, having skin problems makes them more concerned about not having self-care problems.

Backward Group

Perceived impact of bolt-on removal. All backward group participants (*n* = 30) expressed that the removal of bolt-ons made the conditions described in the health states feel less severe. Most indicated that the removal influenced how they interpreted the core dimensions, particularly when vision (*n* = 8; 80%), cognition (*n* = 7; 70%), and social relationships (*n* = 5; 50%) were removed (Table 3). Furthermore, cognition (*n* = 5; 50%), social relationships (*n* = 4; 40%), and vision (*n* = 4; 40%) remained relevant to participants, who continued to consider them in their decisions even after removal.

Shifts in core dimensions' importance due to bolt-on removal. Contrary to our hypothesis, A2 (*k* = 12) generated slightly fewer total shifts than A1 (*k* = 14), with A3 showing similar levels (*k* = 12) (Table 4). The removal of vision (*k* = 8), skin irritation (*k* = 8), and social relationships (*k* = 7) resulted in the greatest number of increases in the perceived importance of core dimensions. Notably, nearly all shifts were increased (*k* = 33, 92%). Decreases in perceived importance were rare, occurring only for vision (*k* = 2) and self-confidence (*k* = 1), with no decreases observed for the other bolt-ons. With the exception of A2, bolt-ons that were removed last (i.e., presented first in the dyad) generated more overall shifts than those removed first: vision (*k* = 10) versus tiredness (*k* = 2), cognition (*k* = 5) versus social relationships (*k* = 7), and skin irritation (*k* = 8) versus self-confidence (*k* = 4). Among the increases (*k* = 33), most were categorized as relative preference (*k* = 19; 58%), followed by measurement (*k* = 8; 24%) and absolute preference (*k* = 6; 18%). Across all arms, perceived importance shifted most frequently in the anxiety/depression dimension (*k* = 11), while the remaining core dimensions registered between 3 and 8 shifts. Illustrative excerpts representing each reasoning category are as follows (more excerpts in Supplementary Materials 3):

Measurement sense. The removal of social relationships increased the importance of anxiety/depression:

I saw social relationships as the key to my psychological state. Having problematic relationships would make my anxiety or depression worse. (A2_B_09, male, 67)

Table 4 Perceived Changes in the Importance of EQ-5D-5L Core Dimensions with Addition or Removal of Bolt-ons^{a,b,c}

Forward Stepwise Group (<i>n</i> = 30 Participants)									
Changes in the Perceived Importance of Core Dimensions after Adding Bolt-ons									
Core Dimension	First Bolt-on Added			Second Bolt-on Added			Total		
	Arm 1	Arm 2	Arm 3	Arm 1	Arm 2	Arm 3	Arm 1	Arm 2	Arm 3
	Vision	Cognition	Skin Irritation	Tiredness	Social Relationships	Self-Confidence	Tiredness	Social Relationships	Self-Confidence
Mobility	3† (M:2, R:1, A:0)	0† (—)	1† (M:1, R:0, A:0)	4† (M:3, R:1, A:0)	1† (M:1, R:0, A:0)	1† (M:0, R:0, A:1)	3† (M:2, R:0, A:1)	1† (M:0, R:0, A:1)	7† (M:5, R:1, A:1)
	1† (M:0, R:1, A:0)	5† (M:0, R:4, A:1)	1† (M:0, R:0, A:1)	7† (M:0, R:5, A:2)	0† (—)	1† (M:0, R:0, A:1)	1† (M:0, R:0, A:1)	1† (M:0, R:0, A:1)	8† (M:0, R:5, A:3)
Self-care	2† (M:2, R:0, A:0)	0† (—)	3† (M:2, R:0, A:1)	5† (M:4, R:0, A:1)	0† (—)	1† (M:0, R:0, A:1)	2† (M:1, R:0, A:1)	1† (M:0, R:0, A:1)	7† (M:5, R:0, A:2)
	2† (M:0, R:2, A:0)	6† (M:0, R:6, A:0)	0† (—)	8† (M:0, R:8, A:0)	0† (—)	1† (M:0, R:1, A:0)	1† (M:0, R:1, A:0)	1† (M:0, R:1, A:0)	9† (M:0, R:9, A:0)
Usual activities	3† (M:3, R:0, A:0)	0† (—)	3† (M:3, R:0, A:0)	6† (M:6, R:0, A:0)	2† (M:2, R:0, A:0)	1† (M:0, R:0, A:1)	3† (M:2, R:0, A:1)	1† (M:0, R:0, A:1)	9† (M:8, R:0, A:1)
	1† (M:1, R:0, A:0)	4† (M:0, R:4, A:0)	0† (—)	5† (M:1, R:4, A:0)	0† (—)	1† (M:0, R:0, A:1)	1† (M:0, R:0, A:1)	1† (M:0, R:0, A:1)	6† (M:1, R:4, A:1)
Pain/discomfort	1† (M:1, R:0, A:0)	0† (—)	4† (M:4, R:0, A:0)	5† (M:5, R:0, A:0)	1† (M:1, R:0, A:0)	1† (M:0, R:0, A:1)	2† (M:1, R:0, A:1)	1† (M:0, R:0, A:1)	7† (M:6, R:0, A:1)
	0† (—)	5† (M:0, R:4, A:1)	1† (M:0, R:0, A:1)	6† (M:0, R:4, A:2)	0† (—)	1† (M:0, R:0, A:1)	1† (M:0, R:0, A:1)	1† (M:0, R:0, A:1)	7† (M:0, R:4, A:3)
Anxiety/depression	3† (M:3, R:0, A:0)	1† (M:1, R:0, A:0)	2† (M:2, R:0, A:0)	6† (M:6, R:0, A:0)	3† (M:3, R:0, A:0)	2† (M:2, R:0, A:0)	5† (M:5, R:0, A:0)	1† (M:1, R:0, A:0)	11† (M:11, R:0, A:0)
	0† (—)	6† (M:6, R:0, A:0)	0† (—)	6† (M:6, R:0, A:0)	0† (—)	1† (M:1, R:0, A:0)	3† (M:3, R:0, A:0)	1† (M:1, R:0, A:0)	9† (M:9, R:0, A:0)
<i>Total</i>	12† (M:10, R:1, A:0)	1† (M:1, R:0, A:0)	13† (M:12, R:0, A:1)	26† (M:24, R:1, A:1)	8† (M:8, R:0, A:0)	1† (M:1, R:0, A:0)	15† (M:11, R:0, A:4)	6† (M:6, R:0, A:0)	41† (M:35, R:1, A:5)
	4† (M:1, R:3, A:0)	26† (M:6, R:18, A:2)	2† (M:0, R:0, A:2)	32† (M:7, R:21, A:4)	0† (—)	1† (M:1, R:0, A:0)	7† (M:3, R:1, A:3)	1† (M:1, R:0, A:0)	39† (M:10, R:22, A:7)
Backward Stepwise Group (<i>n</i> = 30 Participants)									
Changes in the Perceived Importance of Core Dimensions after Removing Bolt-ons									
Core Dimension	Second Bolt-on Removed			First Bolt-on Removed			Total		
	Arm 1	Arm 2	Arm 3	Arm 1	Arm 2	Arm 3	Arm 1	Arm 2	Arm 3
	Vision	Cognition	Skin Irritation	Tiredness	Social Relationships	Self-Confidence	Tiredness	Social Relationships	Self-Confidence
Mobility	3† (M:0, R:3, A:0)	0† (—)	1† (M:0, R:1, A:0)	4† (M:0, R:4, A:0)	2† (M:0, R:2, A:0)	0† (—)	2† (M:0, R:2, A:0)	1† (M:1, R:0, A:0)	6† (M:0, R:6, A:0)
	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	1† (M:1, R:0, A:0)	1† (M:1, R:0, A:0)	1† (M:1, R:0, A:0)	1† (M:1, R:0, A:0)
Self-care	0† (—)	1† (M:0, R:0, A:1)	1† (M:0, R:1, A:0)	2† (M:0, R:1, A:1)	1† (M:0, R:1, A:0)	0† (—)	1† (M:0, R:1, A:0)	1† (M:0, R:1, A:0)	3† (M:0, R:2, A:1)
	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)
Usual activities	3† (M:0, R:1, A:2)	2† (M:0, R:0, A:2)	1† (M:0, R:1, A:0)	6† (M:0, R:2, A:4)	1† (M:0, R:1, A:0)	0† (—)	1† (M:0, R:1, A:0)	1† (M:0, R:1, A:0)	7† (M:0, R:3, A:4)
	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)	0† (—)
Pain/discomfort	1† (M:1, R:0, A:0)	1† (M:0, R:0, A:1)	4† (M:2, R:2, A:0)	6† (M:3, R:2, A:1)	1† (M:0, R:1, A:0)	0† (—)	1† (M:0, R:1, A:0)	1† (M:0, R:1, A:0)	7† (M:3, R:3, A:1)
	1† (M:1, R:0, A:0)	0† (—)	0† (—)	1† (M:1, R:0, A:0)	0† (—)	0† (—)	0† (—)	0† (—)	1† (M:1, R:0, A:0)
Anxiety/depression	1† (M:1, R:0, A:0)	1† (M:1, R:0, A:0)	1† (M:0, R:1, A:0)	3† (M:2, R:1, A:0)	2† (M:2, R:0, A:0)	3† (M:0, R:3, A:0)	7† (M:3, R:4, A:0)	1† (M:1, R:0, A:0)	10† (M:5, R:5, A:0)
	1† (M:1, R:0, A:0)	0† (—)	0† (—)	1† (M:1, R:0, A:0)	0† (—)	0† (—)	0† (—)	0† (—)	1† (M:1, R:0, A:0)
Total	8† (M:2, R:4, A:2)	5† (M:1, R:0, A:4)	8† (M:2, R:6, A:0)	21† (M:5, R:10, A:6)	7† (M:1, R:6, A:0)	3† (M:0, R:3, A:0)	12† (M:3, R:9, A:0)	3† (M:3, R:3, A:0)	33† (M:8, R:19, A:6)
	2† (M:2, R:0, A:0)	0† (—)	0† (—)	2† (M:2, R:0, A:0)	0† (—)	1† (M:1, R:0, A:0)	1† (M:1, R:0, A:0)	1† (M:1, R:0, A:0)	3† (M:3, R:0, A:0)

^aIn the forward stepwise group, participants were asked whether the importance of the original 5 dimensions increased, decreased, or remained unchanged after the addition of the first and second bolt-ons. In the backward stepwise group, participants were asked whether the importance of any of the remaining 5 or 6 dimensions increased, decreased, or remained unchanged after the removal of the seventh and sixth dimensions. Increases and decreases are indicated in this table using † or ‡. Key takeaways:

- Forward group: the first bolt-on added generally generated more shifts than the second one did. Increases in core dimension importance following bolt-on addition were mostly classified as measurement-based reasoning, whereas decreases were more often classified as relative preference.
 - Backward group: the first bolt-on removed generated fewer shifts than the second one did. Increases in core dimension importance following bolt-on removal were mostly classified as measurement-based reasoning, whereas decreases present, measurement based.
 - Across both forward and backward groups, absolute preference was uncommon. Overall, adding bolt-ons prompted greater reevaluation of core dimensions than removal did.
- ^bClassification of perceived importance is presented in parentheses: M = measurement, R = relative preference, A = absolute preference.
- ^cAdding or removing 1 bolt-on may have influenced participants' ratings of multiple core dimensions.

Interpretation: Social relationships viewed as linked to anxiety/depression, reflecting a mental health framing.

Relative preference. The removal of self-confidence increased the importance of anxiety/depression:

With the removal of self-confidence, I focused on depression because I prioritize mental health. (A3_B_02, female, 26)

Interpretation: Removal of self-confidence shifted the focus directly toward anxiety/depression.

Absolute preference. The removal of cognition increased the importance of usual activities:

Thinking is the most important factor in my work, and after it was removed, work became my top priority. This made me think that probably pain is the one that's causing problems in my activities. (A2_B_06, male, 37)

Interpretation: Removing cognition led to a reconsideration of usual activities, as problems previously attributed to the cognition dimension were now seen as problems within the usual activities dimension.

Inference of the level of problems in the bolt-ons. Several participants continued to consider the removed bolt-on(s) during their valuations, inferring their level of problems through the core dimensions (Table 5). This inference was most frequently seen for cognition ($n = 5$), social relationships ($n = 4$), vision ($n = 4$), and tiredness ($n = 3$). All such instances were classified as measurement-based reasoning, reflecting participants' tendency to "fill in" the missing information by drawing links to related dimensions. The inferred severity of problems in the removed bolt-ons was perceived as higher ($k = 9$), approximately the same ($k = 11$), or lower ($k = 3$) compared with the related core dimension. Among the core dimensions, usual activities was most commonly used to infer these levels ($k = 11$), followed by mobility ($k = 7$) and anxiety/depression ($k = 7$). The most frequent pairings, suggesting the strongest conceptual overlap, were usual activities and vision ($k = 4$, "vision is essential for working and other usual activities"), mobility and social relationships ($k = 3$, "limited mobility reduces participation in social activities"), usual activities and cognition ($k = 3$, "thinking ability is essential for solving daily problems"), and anxiety/depression and cognition ($k = 3$, "cognitive processes play a role in emotional regulation and anxiety management").

Discussion

This study is the first to qualitatively explore the thought process of individuals when valuing EQ-5D-5L + bolt-on health states, particularly how bolt-ons interact with the 5 core dimensions. We found that adding or removing bolt-ons often led participants to reconsider the importance of the core EQ-5D-5L dimensions. Cognition and vision generated the most shifts in perceived importance in the forward and backward groups, respectively. However, most of these shifts were not preference based.

The forward group exhibited nearly twice as many shifts as the backward group did, suggesting that participants were more responsive to new additions (and therefore emerging patterns) than to the disappearance of existing features. This was especially notable in forward-ordered A2, where the addition of conceptually related bolt-ons triggered more shifts. Forwarding ordering introduces new information over time, potentially increasing the perceived complexity of the task and prompting more reevaluation of the core dimensions. In contrast, backward ordering removes information and may be processed as an omission or as no longer relevant, resulting in fewer explicit shifts. Different starting points may also create different anchors, with early exposure to bolt-ons driving more shifts than later exposure (ordering/learning effect). Importantly, the nature of participants' reasoning differed by ordering, rendering direct comparisons unsuitable. In the forward group, increases were mostly driven by measurement-based reasoning, while decreases were typically based on relative preferences. In contrast, the backward group showed that increases were mostly driven by relative preferences, and decreases were rare. Across both groups, absolute preference shifts were least common, potentially reflecting that—rather than reassessing their personal health preferences—participants relied on the most cognitively accessible reasoning, such as recognizing conceptual overlaps among dimensions. For example, some perceived skin irritation as a form of pain/discomfort or viewed self-confidence as intertwined with anxiety/depression. Vision—a functioning-specific bolt-on—was often seen as a precondition for everyday functioning (e.g., mobility and usual activities), while tiredness was interpreted as an unavoidable, downstream consequence of problems across other dimensions. Others assumed that having no problems with cognition would help them manage anxiety/depression, believing that thinking ability could regulate emotional health even though the cTTO task specified that no intervention could alleviate

Table 5 Inference of Bolt-on Problem Levels following Removal from EQ-5D-5L Dimensions in the Backward Stepwise Group

Study Arm	Removal Order	Bolt-on	n^a	Core Dimension from Which the Level of Problems Was Inferred							
				Mobility		Self-Care		Usual Activities		Pain/Discomfort	
				Relative Severity in Bolt-on	k	Relative Severity in Bolt-on	k	Relative Severity in Bolt-on	k	Relative Severity in Bolt-on	k
1	1	Tiredness	3	About the same	2	—	—	Higher	1	—	—
	2	Vision	4	Uncertain	1	—	—	Higher	1	—	—
								About the same	1		
								Uncertain	2		
2	1	Social relationships	4	About the same	2	Uncertain	1	Lower	1	Uncertain	1
				Uncertain	1	—	—	Uncertain	1		
	2	Cognition	5	—	—	—	—	Higher	2	About the same	1
								Lower	1		
										Higher	1
										About the same	1
										Lower	1
										Higher	1
3	1	Self-confidence	1	Higher	1	Higher	1	Higher	1	Higher	1
	2	Skin irritation	2	—	—	—	—	—	—	About the same	2
				Higher	1	Higher	1	Higher	4	Higher	1
				About the same	4	About the same	0	About the same	1	About the same	3
				Lower	0	Lower	0	Lower	2	Lower	0
				Uncertain	2	Uncertain	1	Uncertain	3	Uncertain	1
					7		2		11		7
Total											

 k , frequency of inference.^aNumber of participants who still factored in the bolt-on after removal.

Note: All reasonings were all classified as reflecting the measurement sense.

health states over the valuation period. Although all participants completed the cTTO tasks successfully, these patterns suggest that many participants may have forgotten or deprioritized certain aspects of the valuation “game,” a natural human response in complex decision tasks. Most shifts did not reflect tradeoff-based reasoning, highlighting the cognitive difficulty of absolute preference judgments. To preserve authenticity, we deliberately avoided overdirecting participants during interviews. While this yielded organic insights, future studies should consider explicitly probing for absolute preferential shifts, as these are most relevant to valuation outcomes such as years traded. At present, however, the impact of measurement reasoning remains unclear. In the absence of quantitative evidence, we cannot determine whether it influences valuation outcomes or merely redistributes importance without reflecting actual preferences. An alternative interpretation is that some instances classified as measurement reasoning may reflect respondents’ expectations of conceptual coherence in health state descriptions, given perceived linkages between dimensions. Future quantitative research is needed to isolate the effects of different reasoning types. In light of this uncertainty, backward ordering may be considered as a diagnostic tool to assess and potentially reduce measurement-based reasoning, as it elicited fewer measurement-based shifts in our study. Although the forward method yielded more preference-based responses overall, it also resulted in more decrease shifts (contrary to expectations), which may suggest greater difficulty when items are added rather than removed.

Although we observed only a few instances of absolute preferential reasoning, our findings nonetheless raise important considerations for the development and application of the scaling factor model.^{27,28} This model assumes that adding a bolt-on proportionally scales the disutility of core EQ-5D dimensions while preserving their relative importance. However, our findings suggest that this assumption may not hold universally. Participants frequently described interactions between bolt-ons and core dimensions (e.g., linking vision to mobility or cognition to anxiety/depression). While some degree of conceptual overlap is inevitable, these interactions may influence valuation outcomes, particularly when participants engage in preferential reasoning. In some cases, bolt-ons appeared to increase the perceived importance of certain core dimensions, challenging the idea that scaling effects apply uniformly across the descriptive system. This suggests that proportional adjustment may depend on the specific bolt-on used and that a scaling factor

greater than 1 may occasionally be appropriate. Furthermore, some backward group participants inferred higher levels of problems for removed bolt-ons based on problems observed in related core dimensions. This implies that an EQ-5D-5L with a level 1 bolt-on may be valued more highly than EQ-5D-5L alone, as the absence of the bolt-on leaves room for negative inferences. A level 1 bolt-on may thus function as a “pleasant surprise” that provides clarity in the health state description. Our findings also suggest that omitted dimensions may not always be actively considered in participants’ valuation of EQ-5D-5L health states; that is, they may not be consciously interpreted as unspecified or problem free. In such cases, adding a bolt-on does not simply extend the descriptive system but may change how participants interpret and weigh the EQ-5D-5L core dimensions. This highlights the importance of explicitly accounting for reinterpretation effects when designing and modeling bolt-on valuation studies.

Some limitations of our study should be acknowledged. First, the study was conducted in Indonesia. While qualitative research aims to inform rather than generalize, cultural and contextual differences may influence how bolt-on health states are interpreted in other settings. The online mode of administration likely skewed the sample toward urban, higher-income participants. In addition, the overrepresentation of highly educated individuals may have influenced which dimensions were most frequently prioritized. The prominence of cognition in the forward group may partly reflect sample composition. Although prioritization patterns may vary across populations, the identified reasoning types are expected to be thematically transferable beyond this sample. Regarding study design, only levels 2 and 4 were used in the health states; including level 5 (“unable to”) might have yielded different outcomes. The fixed bolt-on order within each arm may have influenced participants’ interpretation. In all subarms except for backward group A2, the first bolt-ons presented (first added in the forward group or last removed in the backward group) prompted more shifts in core dimension importance than the second. This may reflect an ordering effect, in which participants placed greater weight on the first bolt-on encountered. Lastly, it is also important to contextualize the findings in light of differing levels of exposure: participants in the forward group saw the EQ-5D-5L at the start, while those in the backward group had already been exposed to health states with bolt-ons. This discrepancy in familiarity means that, for example, EQ-5D-5L outcomes across groups are not comparable.

Conclusions

This study provides insights into how individuals reason when valuing EQ-5D-5L health states with bolt-ons. Our qualitative findings suggest that interactions occur between bolt-ons and core dimensions and that these interactions depend on the specific bolt-on(s) used. However, caution is warranted when interpreting the effect of bolt-ons on valuation outcomes, as individuals may engage in nonpreferential forms of reasoning. Future qualitative research should explicitly elicit preference-based responses. Additional quantitative work may help distinguish between the effects of measurement-based and preference-based reasoning.

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Ethics

Ethics approval for the study was granted by the Ethics Development Center of Atma Jaya Catholic University of Indonesia (Ref: 008D/III/PPPE.PM.10.05/9/2024).

Consent to Participate

Participants provided written informed consent.

Consent for Publication

Not applicable.

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Availability of Data and Materials

The data used in this study are available from the corresponding author upon reasonable request.

Supplemental Material

Supplementary material for this article is available online at <https://doi.org/10.1177/0272989X261446644>.

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